Rafi Al Mahmud

Software Engineer

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GitHub | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

Computer Science and Engineering graduate with hands-on Python 3 and TypeScript experience developing and maintaining web applications. Built secure REST APIs, React interfaces, automated tests and PostgreSQL-backed workflows. Focused on debugging, clear code, reliable business logic and technical documentation.

TECHNICAL SKILLS

- **Languages:** Python 3, TypeScript, JavaScript (ES6+), SQL, PHP, Java
- **Software Engineering:** Data Structures & Algorithms, OOP, debugging, REST API design, automated testing
- **Frameworks and Databases:** FastAPI, React, Laravel, PostgreSQL, MySQL, SQLAlchemy, Alembic
- **Tools:** Git, GitHub, Docker, Postman, GitHub Actions, pytest, Vitest

PROJECTS

bari: Student Housing and Booking Platform | *Python, FastAPI, React, PostgreSQL* **2026**

- Developed a multi-role rental platform with JWT-secured APIs, React interfaces, property management and user-confirmed bookings; migrated persistence from SQLite to PostgreSQL with Alembic migrations.
- Implemented permission-scoped messaging and booking validation; added backend/frontend regression tests for concurrent conversations and access rules. GitHub | Live demo

MindWell: Employee Wellness Platform | *Python, FastAPI, React, PostgreSQL* **2025 – 2026**

- Built validated REST endpoints with Pydantic, SQLAlchemy models, JWT authentication, protected role-based dashboards and versioned database migrations.
- Containerised backend and frontend workflows with Docker for consistent local and production setup. GitHub

SportEvo: Sports Booking Platform | *React, TypeScript, Laravel, MySQL* **2026**

- Developed sportevo.games with field-specific slot validation, booking overlap prevention, owner approval workflows and commission records.

MangoStudents: Student Digital Magazine | *PHP, Laravel, MySQL* **2025 – 2026**

- Deployed mangostudent.com with relational content models and admin tools that let editors publish articles without manual database changes.

EDUCATION

BRAC University **Sep 2022 – Sep 2026**

B.Sc. in Computer Science and Engineering – CGPA: **3.57 / 4.00** *Dhaka, Bangladesh*

ADDITIONAL EXPERIENCE

Synchropower Engineering | *Data and Estimation Support* **Mar – May 2026**

- Organized spreadsheet data and assisted with estimation worksheets and related analysis over three months.

RESEARCH

WASE: Fog-Robust Object Detection | *Python, PyTorch* **2026**

- Co-authored B.Sc. thesis implementing and evaluating fog-robust object detection; reported an 8–12% mAP@50 gain over baseline. Manuscript in preparation for ICCV submission.

LEADERSHIP AND CERTIFICATION

- **Associate Sub-Body (ASB) Member**, BRAC University Cultural Club (2024 – 2026); coordinated event and club operations.
- **Associate Data Analyst**, DataCamp.